

Eshupriye Belgotra

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Education

Bachelor of Science in Computer Science | University of Victoria

Jan 2023 - Ongoing Victoria, BC

– GPA: 7.38/9.0 ($\approx$ 3.70/4.0) · Coursework: Algorithms & Data Structures, OOP, Software Development Methods, HCI

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Swift

Frameworks & Tools: React, Next.js, Django, Flask, FastAPI, Node.js, REST APIs, Git, GitHub Actions, Linux/Unix, Docker

AI/ML: RAG, vector embeddings, semantic search, NLP, PyTorch, scikit-learn, HistGradientBoosting

Databases & Cloud: PostgreSQL, MySQL, SQL Server, Supabase, Redis, Azure DevOps, SSIS

Experience

Machine Learning Mentee | RBC Borealis – Let’s Solve It

Mar 2026 – May 2026 Victoria, BC

- Analyzed **1,500+ longitudinal youth health records (20+ behavioural features, 3+ years)** alongside 2 senior AI researchers, reducing them to 2D UMAP embeddings that isolated **4+ user cohorts** to inform model design (Python, Pandas, scikit-learn)
- Built and trained PyTorch models to flag at-risk youth navigating chronic illness, learning from 3+ years of longitudinal engagement data to enable earlier clinical intervention
- Designed an analytics dashboard for **4+ clinical and program teams**, visualizing retention curves, engagement trends, and disengagement risk flags with Python, Matplotlib, and GA4

Clinical Application Analyst (Co-op) | Island Health

Sep 2024 – Apr 2025 Victoria, BC

- Designed and built a multi-screen Clinical Governance Membership Management System in Power Apps for **5,000+ members** across BC’s healthcare system, with automated permission workflows, real-time roster edits, and audit-ready compliance reporting
- Engineered automated data pipelines querying and transforming **10,000+ patient records** in SQL Server, reducing manual data-entry time by **80%** and improving cross-system data accuracy
- Migrated legacy mental health housing system from MS Access to SQL Server; designed schema, authored data-loading scripts, and corrected fiscal calculation logic to support **2,000+ annual residential applications**
- Delivered **3 production releases** via Azure DevOps and ServiceNow with **zero critical incidents** through end-to-end QA; authored PowerShell and SSIS automation saving the team **10+ hours per week**

Projects

Numpy Model Server [github.com/eshu3104/numpy-model-server]

Python, Flask, numpy, Docker, pytest · ML prediction server built from raw TCP sockets

Jul 2026

- Created a concurrent HTTP/WSGI server from raw TCP sockets with a bounded thread pool, serving a Flask endpoint that handled **5 simultaneous requests in 0.51s** examined by **3** tests on pytest
- Implemented linear regression from scratch using the numpy normal equation, matching scikit-learn coefficients to within **1e-11**, and packaged the service as a docker image

Open Source – Mozilla Pontoon [github.com/mozilla/pontoon]

Python, Django REST Framework, CSS, pytest · Mozilla’s Firefox localization platform

Jul 2026

- Repaired a UI regression (issue #4239) in a **~120K-LOC** production codebase (**1.7k stars**) by systematically isolating the root cause, shipping a merged, maintainer-reviewed CSS fix (PR #4289) restoring header rendering across all dashboard pages
- Fixed a REST API bug (issue #4182) where renamed projects and locales returned 404s by implementing slug and locale-code history fallbacks across **3** Django REST Framework endpoints, restoring **302-redirect** parity with the web UI, passing **30/30** endpoint tests

Personal Portfolio Site [eshu.earth]

Next.js, Tailwind, Supabase, Redis, Three.js, WebAudio, OpenAI, GitHub Actions

May 2026

- Built a RAG AI chat assistant using OpenAI embeddings and Supabase pgvector, retrieving grounded context with sub-second latency
- Automated knowledge-base sync with a GitHub Actions pipeline that parses the LaTeX resume with Claude, re-embeds with OpenAI, and upserts to Supabase on demand
- Engineered a low-latency TTS pipeline (time-to-first-audio under **500ms**) driving a lip-synced Three.js 3D avatar at **60fps**; hardened the API with per-IP rate limiting, audit logging, and a GDPR-compliant privacy policy

PreCheck.ai [precheckai.tech]

nwHacks 2026 Finalist: Best Use of Gemini API & .tech · 600+ participants

Jan 2026

- Co-led a 4-person team to Finalist among **600+ participants**, owning the system architecture, sprint planning, and the winning pitch for an AI tool scoring insurance claim-denial risk against BC policies (MSP, Pacific Blue Cross, Sun Life)
- Pair-programmed the React/TypeScript frontend and remediation dashboard surfacing the model’s citation-level fix instructions and 0–100 denial probability

SkyNet – Space Biology Knowledge Engine [github.com/eshu3104/nasa-hackathon-2025]

NASA SpaceApps Hackathon 2025

Oct 2025

- Co-built an AI semantic search engine over **608 NASA publications (11,460 chunks)** in Python/Flask, authoring the chunking, OpenAI embedding, FAISS index, and ranking pipeline, with GPT-4o-mini summarization cutting literature-review time by **60%**

Awards

Tenfold Cursor Hackathon 5th Place (2025) · UVicHacks 2026 People’s Choice Award – FoodBridge · AWS DeepRacer 5th/100, within 2s of world record (2024) · UVic Co-op Scholarship (2025)